

Value-Returning Functions

CHAPTER 9

Chapter 9 Topics

- ❖ Program-defined functions
- ❖ Methods for defining program-defined functions
- ❖ Value-Returning Functions
- ❖ Variable scope and lifetimes
- ❖ Passing Information
- ❖ Square Root and Random numbers

Objectives

- ❖ SQRT function
- ❖ Generate Random Numbers
- ❖ Create and Call a function that returns a Value
- ❖ Pass information by value to a function
- ❖ Write a function prototype
- ❖ Understand a variable's scope and lifetime

Functions

A block of code that performs a task.

Every C++ program has at least one function called *main*.



2 Reasons for Functions

- ❖ To avoid duplicating code
- ❖ Breaks large programs to be broken down into small, manageable tasks



Functions are:

- Value-returning or
- Void functions

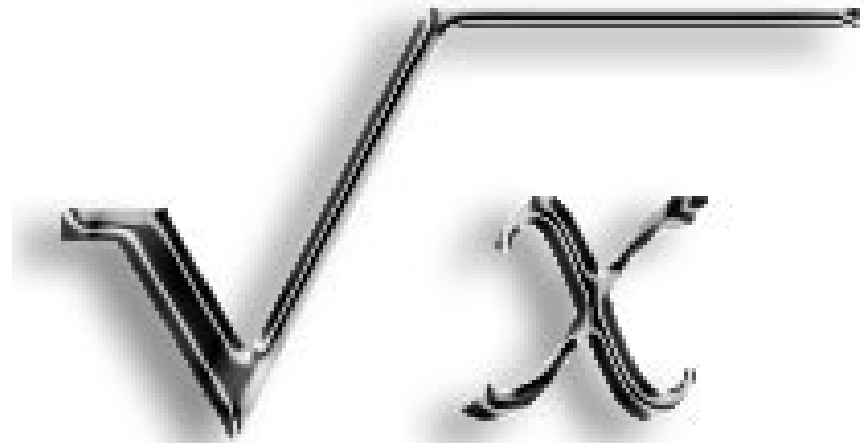


How? Value-returning Functions

- After the task is complete, returns precisely one value.
- Are sometimes built-in:
 - pow
 - toupper
 - tolower
- Others are created by the programmer
 - main

Square Root

- `#include <cmath>`
- `sqrt(x)`
- X must be float or double
- X is an argument



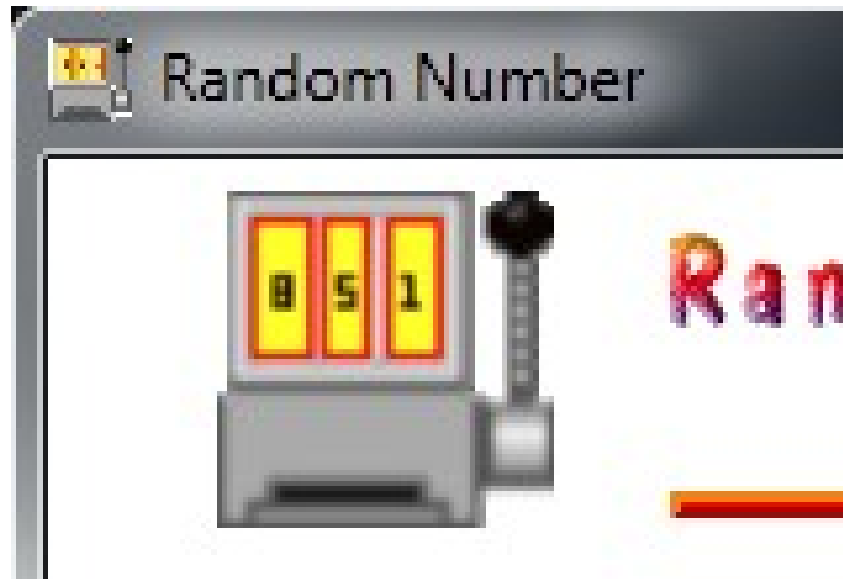
Random Integers



- `rand()` function
- Produces a random number (integer) that is greater than or equal to zero, but
- Less than or equal to the value of `RAND_MAX`.

RAND_MAX

- RAND_MAX is a constant defined in the library file.
- Usually has a value of at least 32,767.



Random Numbers

- Seed (initialize) the random number generator
- Otherwise, you will get the same series of numbers each time the program is executed.
- `srand()` function
- `srand(int time())`



srand() function

- In the cstdlib library file
- `#include <cstdlib>`
- Time function is in the ctime library file
- `#include <ctime>`
- The time function returns the number of seconds that have elapsed since midnight of January 1, 1970.

srand()

Used to generate a set of integers within a given range:

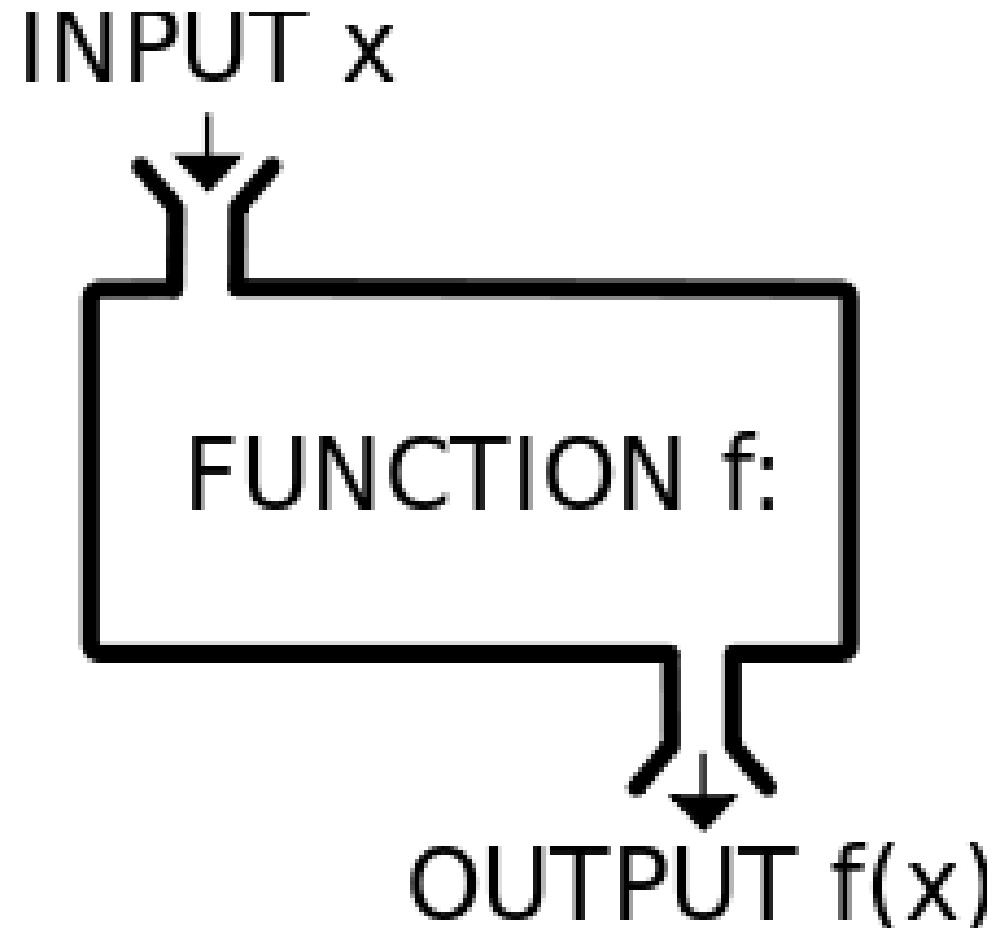
$\text{lowerbound} + \text{rand()} \% (\text{upperbound} - \text{lowerbound} + 1)$

Question 1

A block of code in C++ that performs a specific task is known as _____.

Answer 1

function



Question 2

_____ functions like *main* are written by the programmer.

Answer 2

Program-defined

Question 3

Functions are either void or _____

Answer 3

Value-returning

Question 4

The built-in _____ function can be used to calculate the square root of a number.

Answer 4

sqrt

Question 5

The built-in _____ function can be used to generate random integers within a specific range.

Answer 5

rand

Creating program-defined value-returning functions- Header

- **Code the function header**
 - Defines the type of data the function will return
 - The name of the function
 - Operational parameters
 - `returnDataType functionName ((parameterlist));`

Creating program-defined value-returning functions- Body

Code the function body

- Instructions for accomplishing the task
- Opening and closing braces
- Return statement
- **Return expression**
 - Expression is the value the function returns to the statement that called it
 - Datatypes must agree

Question 6

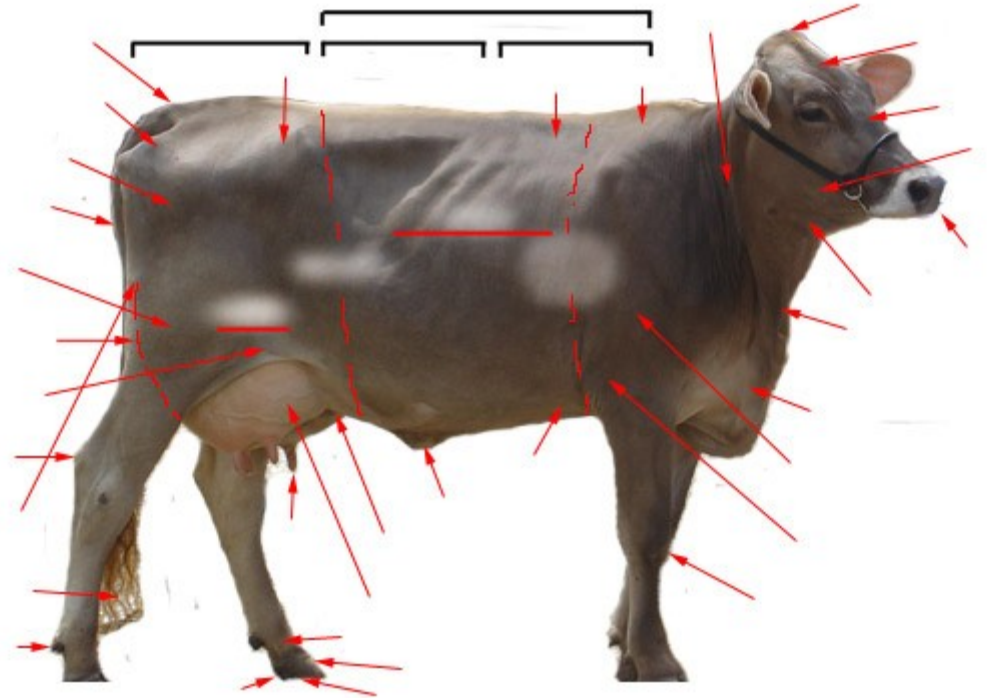
Functions consist of two parts, a _____ and a _____.

Answer 6

Header,



body



Question 7

The optional parameterList in a function header represents the _____ of a function.

Answer 7

Formal arguments or formal parameters of the function. These are the items passed to the function so that it can perform the calculation.

Question 8

The _____ statement of a function returns a value to the calling function.

Answer 8

return

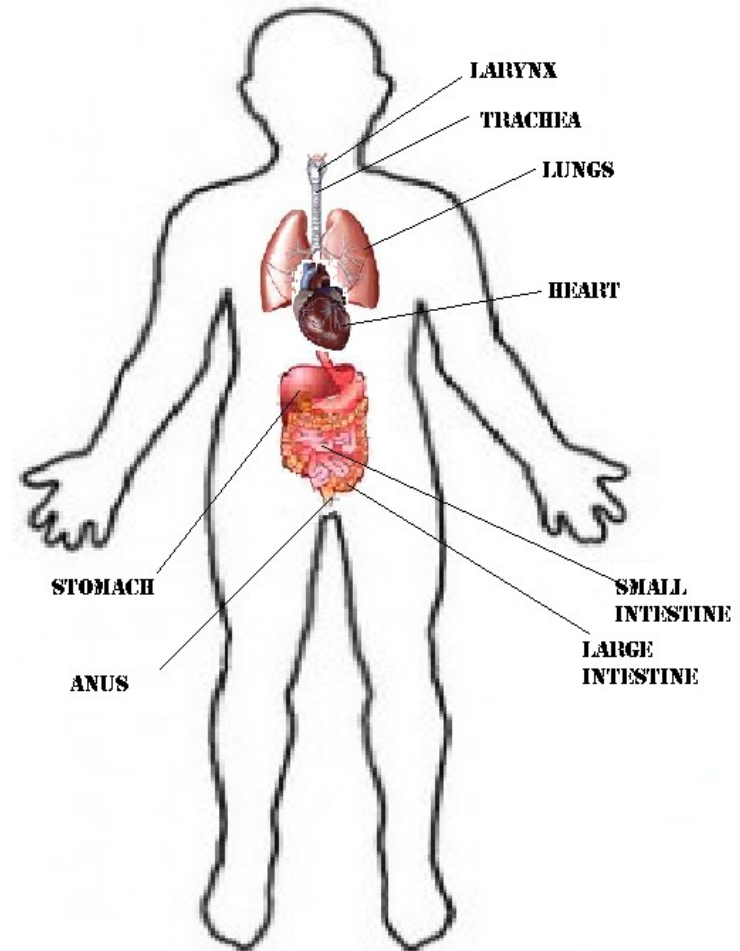


Question 9

The function _____ contains the instructions for the task the function is to perform.

Answer 9

body



Question 10

The return data type of a function is contained in the function _____.

Answer 10

header



Calling a Function

In the function header:

- Parameters or arguments
- Plus datatypes
- You have a relationship between actual and formal parameters
 - Number
 - Type
 - Order
 - Names

Parameters

Actual parameters can be passed to formal parameters as

- Variables
- Named constants
- Literal constants
- Keywords
- Value

Function Prototypes 1

- ❖ Declares a function
- ❖ Contains the datatype of the return value of the function
- ❖ Name of the function
- ❖ The data type of each of the formal parameters

Function Prototypes 2

- ❖ Alerts the compiler that the function will be defined in its entirety later in the program
- ❖ Function is declared before it is called.
- ❖ Ends with a semicolon
- ❖ The function header does not end with a semicolon

Question 11

Another term for calling a function is _____ a function.

Answer 11

invoking



Question 12

The information items passed to a function are called its _____.

Answer 12

Actual arguments or actual parameters

Question 13

Passing a variable's address to a function is referred to as passing by _____.

Answer 13

reference

Question 14

Unless otherwise specified, variables in C++ are automatically passed by _____.

Answer 14

value

Question 15

A function _____ is a statement that specifies the function's name, return data type, and data types of its formal parameters.

Answer 15

prototype



Scope and lifetime of a variable

Scope

- Local
- Global
- Indicates the portion of a program that can use the variable

Lifetime

- Indicates how long the variable will remain in memory

Local

- ❖ Used only by the function in which they are declared
- ❖ Remain in memory until the function ends
- ❖ Makes the function portable (or stand-alone)
- ❖ Could be included in the library for other programs to use



Global

- ❖ Try to avoid using
- ❖ Used anywhere in the program
- ❖ Remain in memory until the program ends
- ❖ Eliminates portability
- ❖ Might require program modifications



Question 16

Variable's _____ indicated where in the program it can be used.

Answer 16

scope



Question 17

A variable's _____ indicates how long the variable remains in the computer's internal memory.

Answer 17

lifetime



Question 18

_____ variables can be used only by the function in which they are declared or in whose parameterList they appear.

Answer 18

local



Question 19

_____ variables are declared outside of any function in the program, and they remain in memory until the program ends.

Answer 19

global



Question 20

Declaring a variable as _____ can allow unintentional errors to occur.

Answer 20

global

